

Strait of Gibraltar



Strait of Gibraltar

Standing as an alluring gateway between ancient Mediterranean civilizations and the mysterious realm beyond, the Strait of Gibraltar has long attracted human interest and investigation. Legend holds that this famous strait was formed by the mighty strength of Hercules, who tore the rocks apart to gain entrance to the Mediterranean. According to this myth, waters from the Atlantic flowed into the Mediterranean with such force that early seafarers were prevented from venturing into the open ocean. It is interesting that this ancient legend does have some basis, in that the present-day surface flow is observed to be dominantly from the Atlantic into the Mediterranean.

The Hercules legend poses an intriguing question: Where does this observed inflow of Atlantic water go? Although a considerable volume is lost by high evaporation within the Mediterranean, this loss is insufficient to offset the volume of inflow. Evaporation causes water in the Mediterranean to become more saline, and thus more dense than the inflowing water. This heavier water sinks and returns to the Atlantic beneath the surface inflow, eventually spreading down across the entire North Atlantic.

This photograph was taken during the flight of Space Shuttle Challenger in October 1984, by Paul Scully-Power at the Naval Underwater Systems Center. It reveals ocean-surface patterns which mirror subsurface events; it has been analyzed by Larry Armi at the Scripps Institution of Oceanography. These patterns are formed by variations in the reflections of sunlight glittering off the sea surface. (See: Spiral Eddies.) The Atlantic is seen to the lower left of the Strait, the Mediterranean to the upper right, Europe above, and Africa below. At the shallowest ridge across the Strait, known as the Camarinal Sill (1), the lower

outflowing layer is accelerated. After passing over the Sill, the current "spills" into deeper water, generating subsurface turbulence which can be detected in the sunglint patterns (2). This phenomenon, termed an internal hydraulic jump, resembles the transition between the smoothly flowing and turbulent water beneath the spillway of a dam.

To the east lies the narrowest distance across the Strait: the Tarifa Narrows (3). Although this section is deeper than over the Camarinal Sill, the inflowing Atlantic water is constricted and thus accelerated, creating internal waves at the interface between the two opposing flows. These waves, reflecting off the walls of the Strait, show up as a cross-hatched pattern in the sunglint (4).

The tidal cycle also has an effect on the flow through the Strait. Tidal currents "slosh" back and forth with the changing tide, driven by a much higher tidal range in the Atlantic than in the Mediterranean. During high tide in the Atlantic, the speed of the outflowing current decreases; at times, it reverses and flows back into the Mediterranean. This reversed flow generates internal waves over the sill, which then propagate eastward. The surface pattern (5) associated with such waves is seen spreading radially from the Gates of Hercules into the Mediterranean: these internal waves were generated several hours earlier at the time of the previous tidal change.

An analysis of the complex flow in the Strait of Gibraltar not only helps improve our understanding of this fascinating region, but can also provide insight into the dynamics of water movement in similar straits. In addition, the Space Shuttle offers a perspective to investigate other subsurface processes through observations of similar surface sunglint signatures.

NASA

National Aeronautics and
Space Administration